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CO1 – AT1 - APPLICATION BASED QUESTIONS

Course Code – DSA0308

Course Name – Natural Language Processing

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Question 1

Answer:

Section 1: Resume Information Extraction Using Regular Expressions

Introduction

In multinational recruitment companies, the HR department receives a large number of resumes daily. Manually reviewing each resume to identify candidate details and technical qualifications is time-consuming and inefficient. Regular Expressions (Regex) provide an automated method for extracting specific information from resume text using predefined patterns.

Explanation

The system uses Python's re module to process resume text and extract important candidate details, including name, email address, mobile number, technical skills, and years of experience. Regular expression patterns are designed to identify each type of information accurately.

The extracted details are organized into a structured candidate profile. The system then evaluates the candidate against the minimum eligibility criteria: at least two years of professional experience and knowledge of Python. Candidates who satisfy both requirements are marked as eligible for shortlisting, while others are marked as not eligible.

Algorithm / Working Procedure

Step 1: Initialization

Import Python's Regular Expression module (re) and prepare the resume text as input for processing.

Step 2: Candidate Identification

Analyze the resume text and extract the candidate's name by identifying the text associated with the name field.

Step 3: Contact Information Extraction

Apply suitable regular expression patterns to identify and extract the candidate's email address and mobile number.

Step 4: Technical Skill Recognition

Define a collection of technical skills, including Python, Java, SQL, Machine Learning, and NLP. Scan the resume and identify which skills are mentioned.

Step 5: Experience Detection

Search the resume for the experience field and extract the number of years of professional experience.

Step 6: Profile Construction

Organize all extracted information into a structured profile containing the candidate's name, contact details, technical skills, and experience.

Step 7: Eligibility Assessment

Evaluate the candidate against the minimum recruitment requirements:

- Professional experience must be at least 2 years.
- Python must be included in the detected technical skills.

Step 8: Report Generation

Display the candidate's complete profile and classify the candidate as *Eligible* or *Not Eligible* based on the assessment.

Outcome: An automated resume screening system that extracts candidate information, evaluates minimum eligibility requirements, and assists the HR department in shortlisting suitable applicants.

Question 2

Answer:

Section 2: Product Search System Using Regular Expressions

Introduction

E-commerce platforms contain large collections of products, making efficient product searching essential for a better customer experience. Traditional search methods may not provide flexible matching for different keyword patterns. Regular Expressions can improve product discovery by supporting exact, prefix, suffix, partial, and case-insensitive searches.

Explanation

The system uses Python's re module to search a predefined list of product names based on user-specified keywords. Different regular expression patterns are applied depending on the selected search method.

- Exact keyword search: Finds products containing the complete keyword as a word.
- Prefix-based search: Finds products containing words that begin with the specified keyword.
- Suffix-based search: Finds products containing words that end with the specified keyword.
- Partial keyword search: Finds products containing the keyword anywhere in their names.
- Case-insensitive search: Matches keywords regardless of uppercase or lowercase letters.

After searching, the system displays all matching product names and calculates the total number of matches for each search operation. The results are organized into a report for easy analysis.

Algorithm / Working Procedure

Step 1: System Initialization

Import the re module and initialize a collection of product names available in the e-commerce database.

Step 2: Search Input Processing

Accept a search keyword and determine the type of search operation to be performed.

Step 3: Exact Keyword Matching

Construct a regular expression using word boundaries to identify products containing the complete search keyword as a word.

Step 4: Prefix-Based Matching

Create a pattern that identifies words beginning with the specified keyword and retrieve the corresponding product names.

Step 5: Suffix-Based Matching

Construct a pattern that identifies words ending with the specified keyword and collect all matching products.

Step 6: Partial Keyword Matching

Search for the specified sequence of characters anywhere within the product names to support flexible keyword searches.

Step 7: Case-Insensitive Processing

Apply the re.IGNORECASE flag to ensure that uppercase and lowercase variations of the same keyword produce consistent results.

Step 8: Result Filtering

Compare the search pattern against every product name and collect all products that satisfy the selected search condition.

Step 9: Match Counting

Calculate the total number of matching products for each search operation.

Step 10: Search Report Generation

Display the search keyword, search method, matching product names, and total number of results in a structured report.

Outcome: A flexible product search engine that supports multiple keyword-matching techniques, improves product discovery, and provides a summary of search results.

Question 3

Answer:

Section 3: University Registration System Using Regular Expressions

Introduction

Universities require accurate student information before permitting course enrollment. Incorrect registration numbers, invalid email addresses, improperly formatted course codes, and invalid mobile numbers can cause registration errors. Regular Expressions provide an effective way to validate student information automatically according to predefined rules.

Explanation

The system uses Python's re module to validate the student's registration details, including register number, institutional email, course code, semester, and mobile number.

A suitable regular expression is defined for each field based on the university's requirements. The register number and course code are checked against their prescribed formats, while the email address is verified against the approved institutional domain. The semester is checked to ensure that it falls within the permitted range, and the mobile number is checked for the correct length and format.

The system displays an individual validation message for each field. After completing all checks, it generates a final registration report. If every required field is valid, the registration is marked as successful. Otherwise, the system reports failure and identifies the fields that need correction.

Algorithm / Working Procedure

Step 1: Registration Initialization

Import Python's re module and collect the student's registration details required for course enrollment.

Step 2: Register Number Verification

Define a regular expression based on the university's register number format and verify whether the submitted register number follows the prescribed structure.

Step 3: Institutional Email Authentication

Validate the email address against the required email format and verify that it belongs to the university's approved email domain.

Step 4: Course Code Verification

Construct a regular expression for the permitted course code format and determine whether the entered course code satisfies the specified pattern.

Step 5: Semester Validation

Check whether the semester input is numeric and falls within the university's permitted semester range.

Step 6: Mobile Number Verification

Apply a suitable regular expression to verify the length and format of the student's mobile number.

Step 7: Field-Level Validation Reporting

Evaluate each field independently and display an appropriate validation message indicating whether the entered information is valid or invalid.

Step 8: Registration Status Evaluation

Combine the validation results of all required fields and determine whether the submitted registration information satisfies every validation rule.

Step 9: Final Registration Report

- If all fields are valid, display **REGISTRATION SUCCESSFUL**.
- If one or more fields are invalid, display **REGISTRATION FAILED** and identify the fields that require correction.

Outcome: A systematic student registration validation mechanism that verifies submitted information, identifies incorrect entries, and determines whether the registration is ready to proceed to course enrollment.

Conclusion

Regular Expressions provide an efficient method for identifying patterns in text and validating structured information. In these three applications, Python automates resume screening, enables flexible product searching, and verifies university registration details. The use of pattern matching, conditional evaluation, and structured reporting improves accuracy, reduces manual effort, and supports faster decision-making.